RETHINKING EVALUATION OF SPARSE AUTOENCODERS THROUGH THE REPRESENTATION OF POLYSEMOUS WORDS

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ABSTRACT

Sparse autoencoders (SAEs) have gained a lot of attention as a promising tool to improve the interpretability of large language models (LLMs) by mapping the complex superposition of *polysemantic* neurons into *monosemantic* features and composing a sparse dictionary of words. However, traditional performance metrics like Mean Squared Error and L_0 sparsity ignore the evaluation of the semantic representational power of SAEs — whether they can acquire interpretable monosemantic features while preserving the semantic relationship of words. For instance, it is not obvious whether a learned sparse feature could distinguish different meanings in one word. In this paper, we propose a suite of evaluations for SAEs to analyze the quality of monosemantic features by focusing on polysemous words. Our findings reveal that SAEs developed to improve the MSE-L_0 Pareto frontier may confuse interpretability, which does not necessarily enhance the extraction of monosemantic features. The analysis of SAEs with polysemous words can also figure out the internal mechanism of LLMs; deeper layers and the Attention module contribute to distinguishing polysemy in a word. Our semantics-focused evaluation offers new insights into the polysemy and the existing SAE objective and contributes to the development of more practical SAEs[1].

1 INTRODUCTION

Sparse autoencoders (SAEs) have garnered significant attention in the field of mechanistic interpretability research (Sharkey et al., 2022; Bricken et al., 2023). SAEs aim to address the *superposition hypothesis* (Elhage et al., 2022), which suggests that activations in large language models (LLMs) can become *polysemantic*[2] — that is, a single neuron corresponds to multiple functionalities — posing a significant challenge for interpretability. By learning to reconstruct LLM activations, SAEs address this challenge by mapping polysemantic neurons to an interpretable space of *monosemantic* features (Sharkey et al., 2022; Bricken et al., 2023). Recent studies have demonstrated the utility of SAEs in various contexts. Templeton et al. (2024) employed SAEs to observe features related to a broad range of safety concerns in Claude 3 Sonnet model, including deception, sycophancy, bias, and dangerous content. Similar investigations have been conducted on GPT-4 (Gao et al., 2024) and Gemma (Lieberum et al., 2024). Moreover, the application of SAEs has extended beyond LLMs to diffusion models (Daujotas, 2024) and reward models (Riggs & Brinkmann, 2024).

SAEs have an architecture consisting of a single, wide layer in the autoencoder, and their training typically involves a reconstruction loss coupled with an L_1 penalty on the features (Sharkey et al., 2022; Bricken et al., 2023). This approach aims to learn features that are both sparse and accurate, creating a known trade-off between Mean Squared Error (MSE) and L_0 sparsity. Previous research has focused on improving this MSE-L_0 Pareto frontier, leading to proposals such as Gated SAE, TopK, and JumpReLU (Rajamanoharan et al., 2024a;b; Gao et al., 2024). Despite improvements in

[1]Code: https://github.com/gouki510/PS-Eval, Dataset: https://huggingface.co/datasets/gouki510/Wic_data_for_SAE-Eval

[2]In this paper, we distinguish *polysemantic/monosemantic* as multiple/single functionalities in the activation of LLMs (Elhage et al., 2022), and *polysemous/monosemous* as multiple/single meanings in the words.

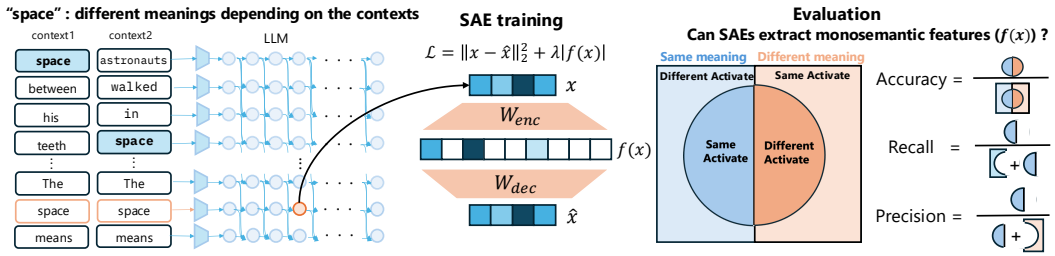

Figure 1: Evaluation of SAE's ability to extract monosemantic features from polysemantic activations in LLM. The word "space" in inputs tokens has different meanings depending on the context: context1 refers to "space" as in the universe (e.g., *...astronauts walked in space...*), while context2 refers to "space" as in a gap or physical distance (e.g., *...space between his teeth...*). SAE is trained by minimizing reconstruction loss and applying a sparsity constraint. The evaluation examines whether the features produced by the SAE can differentiate between the meanings of "space" in different contexts.

the MSE-L_0 Pareto frontier, it serves only as a substitute for the original goal of finding monosemantic features, and we do not evaluate it in practice. Such a gap highlights concerns about evaluation methodologies in SAEs, emphasizing the need for approaches that better align with the objective of discovering meaningful and interpretable features. Motivated by this, prior studies have utilized the IOI task (Wang et al., 2022) with known internal circuits in GPT-2 small (Cunningham et al., 2023; Makelov et al., 2024), RAVEL benchmark with counterfactual labels (Chaudhary & Geiger, 2024), and targeted probe perturbation (Karvonen et al., 2024b), as well as automated assessments using GPT-4 (Cunningham et al., 2023; Karvonen et al., 2024a). However, these metrics are limited to specific models or functional tasks and do not directly assess whether SAE features genuinely represent monosemantic characteristics, which was the original motivation for introducing SAEs.

To address these issues, we evaluate whether SAE features can effectively learn to decompose polysemantic LLM activations into monosemantic features by proposing a metric and dataset, Poly-Semantic Evaluations (PS-Eval). In natural language, polysemantic activations may naturally result from the polysemous words (Li et al., 2024). PS-Eval measures whether SAEs can correctly extract monosemantic features from polysemantic activations (Figure 1), based on the Word-in-Context (WiC) dataset (Pilehvar & Camacho-Collados, 2019), which contains polysemous words whose meanings vary based on context. PS-Eval can assess whether different features in SAE are activated when tokens with different meanings are input, and whether the same features in SAE are activated for tokens with the same meaning. Specifically, we construct a confusion matrix (Table 2) and evaluate the SAE using accuracy, precision, recall, and specificity metrics. To clarify whether our polysemous evaluation helps analyze polysemantic features in SAEs, we employed the logit lens (nostalgebraist, 2020) to examine the impact of monosemantic SAE features on subsequent layers of LLMs. Even in polysemous contexts, these SAE features represent distinct meanings, which indicates that SAEs extract monosemantic features that meaningfully affect the semantic processing in subsequent layers of LLMs.

We conducted experiments to determine which types of SAEs are more conducive to learning monosemantic features. We observed that increasing the dimensionality of SAE latent variables generally leads to better acquisition of monosemantic features, consistent with common scaling practices in SAE research. However, the accuracy saturates as the latent dimension increases, suggesting that there is an optimal latent dimension for practical use of SAE (Templeton et al., 2024; Gao et al., 2024). We also evaluated activation functions such as TopK and JumpReLU, which aim to improve the MSE-L_0 Pareto frontier. However, our results show that these functions do not perform well according to our F1 score metrics. Specifically, the performance ranking is ReLU > JumpReLU > TopK. This indicates that solely optimizing for MSE and L_0 does not evaluate the ability to extract monosemantic features, highlighting the strengths of our proposed evaluations.

Next, we investigate the specific locations in LLMs where SAEs can learn monosemantic features. We first examined how layer depth affects specificity, defined as whether SAE features activate differently for words with different meanings. Our analysis found that deeper layers generally have higher specificity, which indicates that the interpretation of polysemantic tokens occurs gradually as the layers deepen. Also, contrary to the traditional MSE metric, which shows that SAE performance

generally decreases with deeper layers, specificity scores from PS-Eval improve, which highlights that SAE performance cannot be fully assessed using MSE alone. We further evaluated different Transformer components, such as residual layers, MLPs, and self-attention; even within the same layer, the performance of SAEs differed. When examining specificity, attention shows higher values than Residual and MLP. This suggests that even within a single layer of the Transformer, the Attention mechanism contributes to distinguishing polysemous words.

In summary, our primal contributions are:

1. **Introduction of Poly-Semantic Evaluation (PS-Eval)**: We propose a new evaluation metric and dataset, PS-Eval, designed to assess whether SAEs can effectively extract monosemantic features from polysemantic activations. PS-Eval provides a model-independent evaluation metric for the semantic clarity of SAE's features.
2. **Comprehensive Evaluation of SAE**: Our findings suggest that SAEs developed to improve the MSE-L_0 Pareto frontier do not necessarily improve the extraction of monosemantic features. This suggests the need for more holistic evaluation protocols complement to traditional Pareto efficiency.
3. **Insights into Layer Depth and Transformer Components**: Our findings reveal that deeper layers achieve higher specificity and that the attention mechanism contributes to distinguishing polysemous words.

2 Related Works

Sparse Autoencoder Individual neurons in neural networks are polysemantic, representing multiple concepts, and each concept is distributed across many neurons (Smolensky, 1988; Olah et al., 2020; Elhage et al., 2022; Gurnee et al., 2023). SAE is a widely used method that maps these polysemantic neurons to monosemantic high-dimensional neurons, improving the interpretability of activations in neural networks (Sharkey et al., 2022; Bricken et al., 2023). In practice, such polysemantic SAE features have been shown to correspond to diverse semantics, ranging from concrete meanings such as *Golden Gate Bridge* to more abstract concepts like the *entire Hebrew language* (Bricken et al., 2023; Templeton et al., 2024). Recently, SAEs have been applied to large language models such as Claude Sonnet (Templeton et al., 2024), GPT-4 (Gao et al., 2024), and Gemma (Lieberum et al., 2024), where they have been used to discover dangerous content and biased features, as well as to control the behavior of LLMs. SAEs are trained by reconstructing the activations of LLMs and an L_1 sparsity loss of SAE features, which introduces a trade-off between MSE (fidelity) and L_0 (sparsity). In previous research, to improve the MSE-L_0 Pareto frontier, the following approaches have been proposed: scaling the number of SAE features (Templeton et al., 2024; Gao et al., 2024), and introducing better activation functions, such as alternatives to ReLU (Rajamanoharan et al., 2024a; Gao et al., 2024; Rajamanoharan et al., 2024b).

One of the challenges with SAEs is the difficulty of evaluation. While MSE and L_1 sparsity are imposed during training, it is uncertain whether improving this Pareto optimality actually leads to the acquisition of monosemantic features. Existing research has used evaluations like the IOI task in GPT-2 (Makelov et al., 2024; Cunningham et al., 2023), where internal circuits are well understood, or automated evaluations using GPT-4 (Cunningham et al., 2023; Karvonen et al., 2024a). However, there is currently no direct evaluation metric to assess whether SAEs achieve their primary goal of mapping polysemantic activations into monosemantic spaces. In this study, we evaluate how well SAEs can acquire monosemantic representations at the level of individual word meanings using the Word in Context dataset (Pilehvar & Camacho-Collados, 2019), which includes polysemous tokens with meanings that vary depending on context.

Context Sensitive Representations In natural language processing, word embeddings have been limited by their ability to represent only static features, where each word is associated with a single embedding regardless of the context in which it appears. To address this limitation, the Word-in-Context (WiC) dataset (Pilehvar & Camacho-Collados, 2019) was proposed for the evaluation of context-sensitive word embeddings. Currently, a multilingual version of WiC (Raganato et al., 2020) has been developed, and it has also become a part of the widely-used SuperGLUE benchmark (Wang et al., 2020). Because polysemantic features may naturally come from the polysemous words (Li